AI-Assisted Development Record

for *One complex metric and geometric adelic nefness*

A supplementary record of mathematical drafting, checking and revision
Reference: the author-supplied LaTeX source received on 6 November 2026

1 Purpose and reference manuscript

This supplement documents AI-assisted mathematical drafting, checking and revision for the paper. It maps the retained development materials to the present proof, distinguishing early exploration, later manuscript content and subsequent checks. Its purpose is to document assistance and surviving records; it supplies no independent mathematical certification.

The reference is the unmodified `main.tex` and `references.bib` in the latest source ZIP supplied by the author. The separately supplied PDF differs in its abstract and theorem presentation. The archive therefore provides a PDF rebuilt from the exact source and records separate fingerprints for the two inputs.

The source already contains a one-sentence generative-AI disclosure. A more informative replacement paragraph is given in Section 5 and as a separate insertable LaTeX file. The reference manuscript is unchanged.

2 Forms of assistance in the retained record

The documented assistance includes substantial mathematical drafting as well as editorial work:

- preparing candidate arguments for approximation and positivity, with explicit hypotheses, fixed auxiliary data and boundary estimates;
- exploring a prescribed-metric Bergman route and constructing counterexamples to particular intermediate shortcuts;
- locating and comparing mathematical sources, and checking the scope of inputs such as multiplier ideals and global generation;
- revising proofs, definitions, source conventions and exposition in response to visible requests and recorded criticism.

The visible human contributions include choosing directions, adopting or correcting proposals, clarifying the ultimate goal, and requesting greater rigor and a clearer arrangement of estimates. H01–H02 and H10–H17 preserve these requests and replies. A quoted model proposal and the decision to pursue it remain distinct acts; no percentage division of credit is inferred.

The earliest complete initiating exchange and the first message triggering the shorter geometric proof remain unavailable. Unobserved decisions are not reconstructed.

3 Correspondence with the present proof

The present manuscript obtains geometric nefness through semiample approximation. It begins with compatible model and metric data, extends across the boundary, controls multiplier ideals, principalizes the *image ideal*, and obtains exact global generation after ample perturbation. Boundary control gives the fixed-auxiliary nefness conclusion. The number-field argument then uses finite algebraic specialization and Galois-conjugate tensor descent.

Map	Work represented in the current source	Number, PDF page
M01	Compatible boundary presentations and psh extension	3.2, p. 9
M02	Ample adjustment with the specified open restriction	3.3, p. 10
M03	Multiplier-ideal sandwich and image-ideal principalization	3.4, p. 11
M04	Full Cartier bound without assuming flat pullback of the ideal	3.5, p. 11
M05	Exact global generation after an ample perturbation	3.7, p. 12
M06	Boundary error and one fixed ample witness	(3.11), p. 14
M07	Finite algebraic specialization and Galois descent	4.2, p. 14
M08	Current Gross-Schoen object and curvature input	5.1, p. 17

Table 1: Locations are taken from the reference source and its compiled cross-references. The complete map also records source lines, historical H-records and the strength of each correspondence.

For example, the current proof uses a corrected extension B_t and the inclusions

$$\mathcal{O}(-2mtD_t) \subseteq \mathcal{I}_{t,m} \subseteq \mathcal{O}.$$

After principalization,

$$\mathcal{I}_{t,m}\mathcal{O}_Y = \mathcal{O}_Y(-F), \qquad 0 \leq F/m \leq 2t\rho^* D_t.$$

The resulting semiample line

$$N = \rho^*(B_t + \varepsilon A_t) - F/m$$

approximates the prescribed geometric object in the boundary norm. The map links this content to retained drafts and checking notes. Structural correspondence alone establishes neither receipt of a particular input nor the cause of an edit.

Classical tools cited by the manuscript remain credited to their sources. The assistance record does not assign Nadel vanishing, regularity, GAGA, principalization or the adelic definitions to the AI workflow.

Earlier exploration versus the argument retained

The C003/C005/C006 material treats a stronger specified-metric task: boundary convergence of a prescribed global Bergman quotient sequence. H06–H09 and H18–H24 retain the task, candidate, inherited construction, source comparison, countertest and AI review. The countertest invalidates an original-coordinate oscillation shortcut; the candidate uses an adapted model and retains its Jacobian instead.

The present geometric proof uses multiplier-ideal bounds and a semiample model construction. It does not require convergence of these quotient metrics to the prescribed metric. The earlier route is preserved as development history.

The retained ADD-001 manuscript contains a signed object-comparison lemma. The latest source states the Gao–Zhang construction and uses the height/curvature input directly. The older lemma and its checking record are historical material, with no automatic transfer of review status to the current application.

4 Development and revision records

Dates below identify retained records or their stated scope, rather than the first discovery time of an idea.

Both the countertest and the candidate's explicit response survive. The exact pre-feedback draft and authenticated receiving message remain missing, so the record does not establish the full adjacent repair chain.

The earlier internal audit retains its ADD-001 target and its checks of specific obligations. These AI checking statements are not an independent human referee report.

Stage	Retained work	Entry records
23 September	Approximation plan, the specified-metric gap, R002 argument and R003 norm/fixed-auxiliary work	H01–H05, H25–H26
24 September	General local task, adapted-model candidate, countertest and internal validation summary	H06–H09, H18–H24
2–4 October	Requests about rigor and length, the three-lemma organization, and upper/lower estimate presentation	H13–H17
Retained geometric drafts	Baseline, completed-order revision and later comparison/sign revision	Earlier manuscript sources
Latest supplied version	Current compactified-metric formulation, geometric proof, descent and direct application	Reference source

Table 2: A compressed development guide, not a complete chat chronology. The archive's record index states the evidential kind of each item.

5 Suggested manuscript disclosure

The following paragraph expands the existing AI-use sentence while remaining within the documented scope:

> Generative AI tools were used during the development and revision of this work, including the preparation of draft mathematical arguments, the exploration of alternative proof routes and counterexamples, assistance with locating and comparing references, and the organization and revision of the exposition. A supplementary development archive records the retained materials and their relationship to this manuscript, distinguishing earlier explorations from the arguments retained in the present proof. The archived AI-generated drafts and review statements do not constitute independent mathematical certification.

The author-review text is also provided in `AI_ASSISTANCE_EN.txt` and `AI_ASSISTANCE_EN.tex`.

6 Archive guide and evidential limits

The archive contains the exact latest source and matching rebuilt PDF, three earlier source snapshots, selected work records, earlier checking notes, a version diff and the process map. Historical materials and current editorial products are identified separately.

This selected public archive retains visible mathematical arguments and criticism. It excludes private account identifiers, operational logs, system messages, credentials and hidden internal reasoning. It is not a complete native conversation export; distinctive content may identify the project.

Hashes, compilation, link checks and visual inspection verify delivery integrity, not mathematical correctness, historical priority or missing events.

Reading and verification. Start with the repository README, then the process map and record index. The offline verifier checks hashes, relative links and source-label bindings. The paper remains the reference for mathematical statements and proofs.